

Geometric Root Calculators

Fixed circles, global convergence, and analog realization

Ivan Besevic¹ · Independent research

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<https://github.com/ivan-fr/pandrosion>

Abstract. An iteration formula does not specify how to construct its next value. We study explicit line-and-circle realizations for positive roots. A rationally prepared fixed circle acts as a quadratic solver for an inverse Padé correction: the cubic example uses eight mobile joins, and the general linear-power protocol uses $2p + 2$. We prove strict logarithmic-error contraction on its entire real inverse branch for every integer $p \geq 3$; invariant-band convergence follows, with local order five. Centered AD supplies entry from arbitrary positive starts and handles excluded finite charts. A binary-power realization counts this combined protocol without claiming a global eight-line bound. Five Pandrosion constructions, including linear MA and the globally monotone decentered fifth-order arc, locate the result among explicit geometric alternatives. Finally, we explain how the geometry specifies arithmetic cells, branch selection and state storage for analog design, distinguishing a proposed centered quadratic cell from the existing order-three AD behavioral prototype. Incidence certificates, Lean audits, numerical work tables and executable geometry accompany the paper.

1 Why construct an iteration?

A numerical algorithm names operations. A physical calculator must realize them: a ratio may be an alignment, a product a similar-triangle construction, and a square root a circle crossing. Two formulas with the same local order can therefore have different realization costs. This paper asks which geometric arrangements make a root iteration economical and how that arrangement can guide an analog implementation.

The intended setting is constructive computation: exact incidence geometry supplies a design and a correctness argument; a count of mobile operations describes the protocol. It also exposes which quantities must be stored, copied or solved for in a circuit. Drawing cost and electrical delay are different quantities, but the decomposition into reusable operations connects them.

Our principal example is a fixed circle that solves for a correction rather than merely transferring a length. Three joins after the power stage complete a fifth-order update. The circle converges throughout its real branch; a globally defined AD fallback handles entry and geometric exceptions. The decentered arc gives another choice: global monotone convergence with a moving radius. The contribution is these explicit realizations and their analysis, rather than the classical Newton, Halley or Padé families. Section 8 collects the proof and implementation boundaries; the main text develops the constructions themselves.

¹Computational assistance: AI was used for derivations, proof development, figures and manuscript preparation. The author is responsible for the claims and their attribution. Section 8 distinguishes exact proofs, numerical checks and exploratory displays.

2 Coordinates as reusable computing elements

Let $X > 0$, let $p \geq 3$ be an integer, and write

$$a = X^{-1/p}, \quad t = Xs^p, \quad s^+ = s\Phi(t), \quad s > 0.$$

The state approaches the reciprocal root a ; inversion supplies the desired $X^{1/p}$. We use $e = \log(s/a)$ for local error. Points are capital letters, scalar iteration maps are $\mathcal{T}_{\text{label}}$, and polynomial coefficients use Greek letters. Thus a point and a correction function never share the same symbol.

In a rectangle with vertices $C = (0, 0)$, $B = (W, 0)$, $O = (0, H)$ and $A = (W, H)$, define two extended rails and a family of top centers:

$$R(q) = (W, H(1 - q)), \quad L(q) = (0, H(1 - q)), \quad \mathcal{U}_f = \left(\frac{Wf}{f-1}, H \right). \quad (1)$$

A line through the prepared center performs multiplication in one direction and division in the other:

$$\mathcal{U}_f R(q) \cap OC = L(fq), \quad \mathcal{U}_f L(b) \cap AB = R(b/f). \quad (2)$$

For example, substituting the first two points into a line equation and setting $x = 0$ gives ordinate $H(1 - fq)$. This is the whole arithmetic content of the rail device.

2.1 The power stage

Start at $P = R(s)$. The join CP meets the top line at $V = \mathcal{U}_{1/s}$. A leftward projection through $\mathcal{U}_{1/2}$ followed by a return through V changes a rail value q to $sq/2$. After $p - 1$ pairs, the endpoint is $Q = R(s^p/2^{p-1})$. Including the initial join, the stage uses $2p - 1$ lines. Its normalization factor is part of the design, not discarded arithmetic.

More generally, choosing the leftward factors $f_1, \dots, f_{p-1}$ gives an endpoint with value $s^p \prod_i f_i$. This allows the power stage to feed a device with its preferred residual scale. We use exactly this freedom for the fixed-circle calculator.

Power protocols, one state encoding. The later rectangle drawings use a different power stage: similar triangles and parallels produce $E = R(s^p)$ directly. They do not change the meaning of $R(s)$. The rail telescope instead produces a scaled power; its scale is absorbed into the prepared readout. Their costs must be kept separate.

Power stage	Geometric mechanism	Output and use
Rail telescope	$2p - 1$ joins; repeated projection	Scaled s^p ; pencils and fixed circle
Triangle chain	Horizontals and similar-triangle parallels	$R(s^p)$; original MA/AK/AD drawings
Binary products	Repeated squaring and multiplication	$R(s^p)$; large degrees and safe controllers

2.2 A precise notion of work

We count a join as J , a parallel-construction macro as $P_{\parallel}$, and a mobile compass arc as C_{arc} . Intersections of already drawn objects, fixed preparation and final inversion are excluded. When expanded into ruler-and-compass traces, our parallel macro costs $2J + 3C_{\text{arc}}$. These conventions are specified before any cost comparison.

This question belongs to the constructive tradition of Lemoine's géométrie and d'Ocagne's nomography [1, 2]. The Poncelet–Steiner theorem supplies a broad existence result for straightedge constructions with a given circle and its center [3]. Our question is more specific: an explicit repeated protocol, its branch selection, and its mobile cost. The name Pandrosion refers to the addressee of the opening of Book III of Pappus' *Collection* [4]. Our rectangle follows a modern reconstruction [5]; the accelerated supports are modern constructions, not attributed to the ancient text.

3 A prepared circle as a quadratic solver

Suppose the desired correction factor is $v \approx 1$. Instead of evaluating a rational approximation to $t^{-1/p}$, approximate the inverse relation $t = v^{-p}$ by a quadratic rational model and solve it exactly. Put

$$\begin{aligned} \alpha_p &= (p+1)(p+2), & \beta_p &= p^2 - 4, & \gamma_p &= (p-1)(p-2), \\ N(v) &= \gamma_p v^2 - 2\beta_p v + \alpha_p, & D(v) &= \alpha_p v^2 - 2\beta_p v + \gamma_p. \end{aligned} \quad (3)$$

The model is $\mathcal{R}(v) = N(v)/D(v)$, the $[2/2]$ Padé approximant of v^{-p} at one. Its inverse branch $v(1) = 1$ defines $\mathcal{T}_{\text{circle}}(s) = sv(Xs^p)$. A circle can supply the quadratic root in a single crossing.

3.1 The eight-line cubic example

Use $W = 2, H = 4, p = 3$ and $X = 2$. Prepare a circle through B with center O_Γ , a projector Z , and constants

$$O_\Gamma = (-152/113, 126/113), \quad Z = (478/113, 396/113), \quad \rho = \frac{1}{2}, \quad k = \frac{5}{113}.$$

The first five joins form the rail power stage with left factors ρ and $\sigma = 20/113$, retaining $L_1 = L(\rho s)$ and ending at $T = R(kXs^3)$. The last three joins are the main device:

1. Draw ZT and select its prescribed intersection G with the fixed circle.
2. Draw BG to the top line, obtaining $U = \mathcal{U}_{\rho/v}$.
3. Draw UL_1 back to the right rail. Its value is $(\rho s)/(\rho/v) = sv$.

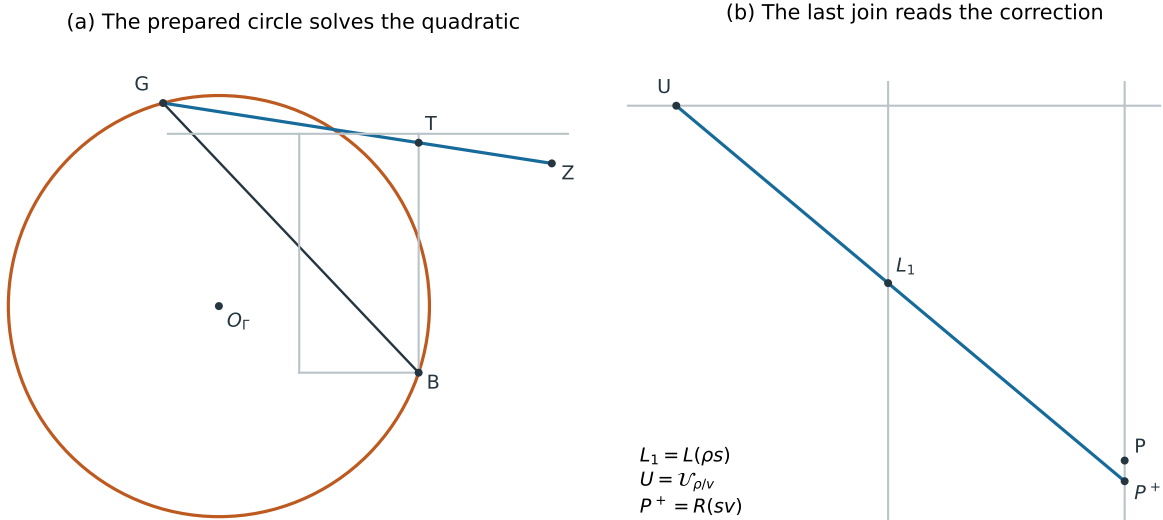


Figure 1: The fixed-circle device at $s = 3/4$. Left: the sixth join supplies G , and the seventh supplies the readout center U . Right: an independently enlarged view shows the eighth join and distinguishes P from P^+ . The circle is drawn once, before iteration. Equal horizontal and vertical scales are used within each view.

The computed next state is $0.793700551706633 \dots$, with relative error about 3.24×10^{-8} . The circle is doing arithmetic: it replaces a longer rational readout by the solution of a quadratic model.

3.2 Why the inverse correction has order five

The local order follows from one polynomial identity, which is also a useful certificate for the approximation. At $v = 1$, $N = D = 12$, and

$$pND + v(N'D - ND') = p\alpha_p\gamma_p(v-1)^4. \quad (4)$$

Moreover $\gamma_p N = (\gamma_p v - \beta_p)^2 + 3(p^2 - 4)$ and $\alpha_p D = (\alpha_p v - \beta_p)^2 + 3(p^2 - 4)$, so the logarithms used below are well defined for $p > 2$.

Theorem 3.1 (Local inverse-circle order). *For real $p > 2$, the inverse branch through $v = 1$ exists near $t = 1$ and satisfies*

$$e^+ = -\frac{(p^2 - 1)(p^2 - 4)}{720}e^5 + O(e^7). \quad (5)$$

It therefore alternates sides locally and has exact order five.

Proof. Set $v = 1 + u$, $E(u) = \log \mathcal{R}(1 + u)/p$ and $K(u) = E(u) + \log(1 + u)$. Equation (4) gives

$$K'(u) = \frac{\alpha_p\gamma_p u^4}{(1+u)N(1+u)D(1+u)}.$$

Thus $K(u)/u^5 \rightarrow \alpha_p\gamma_p/720$, while $E(0) = 0$ and $E'(0) = -1$. The inverse function theorem identifies $E(u)$ with the input logarithmic error. Dividing the limits gives the coefficient in (5). Analyticity and $\mathcal{R}(1/v) = 1/\mathcal{R}(v)$ make the logarithmic update odd, giving the stated remainder. The coefficient is nonzero for $p > 2$. $\square$

The same identity, integrated as the derivative of $\log(v^p \mathcal{R}(v))$, proves the fifth-order Padé contact. Model-based iteration and quadratic approximants have classical precedents [6–8]; the construction problem is to realize this inverse with a short, prepared geometric device.

3.3 Which circle intersection?

The discriminant after removing a factor four is

$$\Delta_p(t) = (42p^2 - 24)t - 3(p^2 - 4)(1 + t^2).$$

The transverse real interval is $\tau_- < t < \tau_+$, where

$$\tau_{\pm} = \frac{7p^2 - 4 \pm 4p\sqrt{3(p^2 - 1)}}{p^2 - 4}, \quad \tau_- \tau_+ = 1.$$

Select the descending branch $\mathcal{R}'(v) < 0$, equivalently

$$v_- < v < v_+, \quad v_{\pm} = \frac{p^2 + 2 \pm \sqrt{12(p^2 - 1)}}{p^2 - 4}.$$

Its reference point is $v = 1$. At $p = 3$, $t = 0.05$, both inverse roots are positive, approximately 3.1183 and 6.3817; only the first is on this branch. Positivity alone cannot choose G . At a transverse endpoint the two roots merge. Theorem 3.3 below proves strict error contraction on this entire interval. Tangencies remain excluded from the finite protocol, and the real interval is not the whole positive axis.

3.4 The general-degree construction

For each integer $p \geq 3$, prepare $O_\Gamma = (h, j)$, $Z = (z_x, z_y)$, $k > 0$ and $\rho > 0$, using the rational formulas in Appendix A. The circle passes through $B = (2, 0)$ and is independent of X and s . Put $\sigma = kX2^{p-3}/\rho$. The rail sequence uses

$$f_1 = \rho, \quad f_i = \frac{1}{2} \quad (2 \leq i \leq p-2), \quad f_{p-1} = \sigma.$$

It saves $L_1 = L(\rho s)$ and ends at $T = R(kt)$, because $\prod_i f_i = kX$. The same three final joins then give $R(sv)$.

Theorem 3.2 (Counted finite-circle protocol). *For the preparation in Appendix A, assume $X, s > 0$, $\tau_- < Xs^p < \tau_+$, and*

$$s \neq 1, \quad \sigma \neq 1, \quad v \neq \rho, \quad b(v) = (2-h)2v - 4j(\rho - v) \neq 0.$$

The selected branch is realized by $2p + 2$ mobile joins using the prepared circle.

Proof. The power stage costs $1 + 2(p-1) = 2p-1$ joins. To certify the remaining incidence, write

$$d = (2v, 4(\rho - v)), \quad n = d \cdot d, \quad b = (B - O_\Gamma) \cdot d, \quad G = B - \frac{2b}{n}d.$$

The pencil line through B has direction d . Substituting $B + \xi d$ into the circle equation gives $\xi(\xi n + 2b) = 0$: besides B , the other intersection is exactly the displayed G . Thus $b \neq 0$ keeps the two points distinct. The same line meets the top at $\mathcal{U}_{\rho/v}$ when $v \neq \rho$.

It remains to show that the projector Z and the residual point T select this G . Three points are collinear exactly when the determinant of their displacement vectors vanishes. Multiplying that determinant by n gives

$$I = (2 - z_x)(-2bd_y - z_y n) - [4(1 - kt) - z_y][(2 - z_x)n - 2bd_x].$$

The prepared parameters satisfy $I = \lambda_p[N(v) - tD(v)]$, with $\lambda_p \neq 0$. This identity converts the geometric incidence into exactly the inverse model, rather than an approximation to it. Since the selected root has $N(v) - tD(v) = 0$, it places G on ZT . Projecting G through B , then L_1 through the resulting center, gives $R(sv)$ by (2). These are three joins. The branch arc is parameterized by $v_- < v < v_+$ and can be marked during preparation using its endpoints and $G(1)$. The listed hypotheses exclude an infinite center, a coincident B, G , and tangency; the other rail intersections are transverse. $\square$

The exclusions concern this finite chart. In particular, $s = 1$ sends the initial center V to infinity even though the scalar correction can remain regular. A positive rescaling $\tilde{X} = c^p X$, $\tilde{s} = s/c$ preserves t and conjugates every correction. A dyadic choice can keep the limiting state away from one, controlling the distant initial center near convergence. The circle-specific exclusions still require their own branch and chart checks.

The result is a concrete upper bound on mobile work, not merely an existence theorem for straightedge construction. Its advantage over the rational fifth-order baseline will be quantified after the other devices have been introduced.

3.5 Strict error contraction on the whole real branch

The finite real interval is a convergence region, not merely a domain on which a square root exists. The result below closes the whole-branch conjecture stated in V20. “Contraction” here means strict reduction of the error relative to the root; it does not assert a common Lipschitz constant between arbitrary states.

Theorem 3.3 (Whole-branch contraction). *Let $p \geq 3$ be an integer and $\tau_- < t < \tau_+$, $t \neq 1$. For the descending inverse branch $v = v(t)$,*

$$\left| \frac{\log t}{p} + \log v \right| < \frac{|\log t|}{p}.$$

Consequently the scalar circle iteration stays inside this residual band and converges to $X^{-1/p}$ from every start in it. Its local order is five.

Proof. Reduce the error comparison to a sign. Parameterize the branch by v , so $t = N(v)/D(v)$. Put

$$h(v) = p \log v + 2 \log \frac{N(v)}{D(v)}, \quad S(v) = 2\alpha_p \gamma_p (v-1)^4 - N(v)D(v).$$

The difference of squared errors is $\log v h(v)/p$. It suffices to prove $h(v) < 0$ for $1 < v < v_+$; reciprocity gives $h(1/v) = -h(v)$ and handles the other half. Equivalently, for integer p the desired sign is $(v-1)[D(v)^2 - v^p N(v)^2] > 0$.

Control the shape of h . The certificate (4) gives

$$h'(v) = \frac{pS(v)}{vN(v)D(v)}.$$

Write $v = 1 + u$. The useful simplification is

$$\frac{S(1+u)}{u^2} = \alpha_p \gamma_p u^2 + 12(p^2 - 4)u + 12(p^2 - 16) - \frac{288}{u} - \frac{144}{u^2}.$$

For $u > 0$, its derivative is strictly positive. Since $S(1) = -144$, h first decreases and can then increase only once. Thus $h(1) = 0$ and the single endpoint inequality $h(v_+) < 0$ imply $h(v) < 0$ throughout $(1, v_+)$. The next page proves that endpoint inequality for every integer degree.

Pass from one step to convergence. Let $z = \log t$ and $F(z) = \log[tv(t)^p]$. The inverse branch is continuous in the open band, $F(0) = 0$, and $|F(z)| < |z|$ for $z \neq 0$. Since $\tau_- \tau_+ = 1$, an orbit starting at z_0 remains in the compact interval $[-|z_0|, |z_0|]$ strictly inside the band. Its absolute errors decrease to a limit ℓ . If $\ell > 0$, a convergent subsequence has limit z_* with $|z_*| = \ell$; continuity would give both $|F(z_*)| = \ell$ and $|F(z_*)| < \ell$. Hence $\ell = 0$. The local order follows from (5). $\square$

The first, strict-error assertion is `whole_branch_contraction` in `RectangleFixedCircleBranch`; its 33-declaration audit checks the sign and endpoint argument. The invariant-band convergence and inverse-branch assembly above are written corollaries. This distinction avoids calling the entire analytic/geometric program a single formal theorem.

3.6 An endpoint certificate valid at every degree

Let $r = \sqrt{12(p^2 - 1)}$, $u_+ = (6 + r)/(p^2 - 4)$ and $v_+ = 1 + u_+$. At this branch endpoint,

$$(p^2 - 4)u_+^2 = 12(u_+ + 1), \quad \frac{N(v_+)}{D(v_+)} = \tau_-.$$

Therefore $h(v_+) < 0$ is exactly $v_+^p < \tau_-^2$. The quadratic relation reduces the polynomials to linear expressions:

$$\begin{aligned} N(1 + u_+) &= \frac{6[2(2p + 1) - (p^2 - 2p - 2)u_+]}{p + 2}, \\ D(1 + u_+) &= \frac{6[2(2p - 1) + (p^2 + 2p - 2)u_+]}{p - 2}. \end{aligned}$$

For $3 \leq p \leq 9$, rational brackets for r give exact rational upper bounds on $v_+^p N(v_+)^2 / D(v_+)^2$ below one:

p	3	4	5	6	7	8	9
Lower r	9.79	13.41	16.97	20.49	24	27.49	30.98
Upper r	9.80	13.42	16.98	20.50	24.01	27.50	30.99
Ratio $<$	.130	.147	.154	.157	.159	.161	.162

The displayed decimal endpoints are exact rationals; the ratio bounds are rounded upward. The checked substitution uses the upper bound for v_+ , and the lower bound for u_+ in both reduced expressions, bounding N upward and D downward.

For $p \geq 10$, $3.46p - 1 \leq r \leq 3.47p$ implies $pu_+ \leq 5$ and $D(v_+) \geq 12.3N(v_+)$. For the latter inequality, clearing positive denominators leaves the certificate

$$409q^4 + 11314q^3 + 140038q^2 + 1027656q + 3519360 > 0, \quad q = p - 10 \geq 0.$$

Hence $v_+^p \leq e^{pu_+} \leq e^5 < 148.42 < 12.3^2 \leq [D(v_+)/N(v_+)]^2$. The finitely many rational certificates and this exponential estimate cover all integer $p \geq 3$.

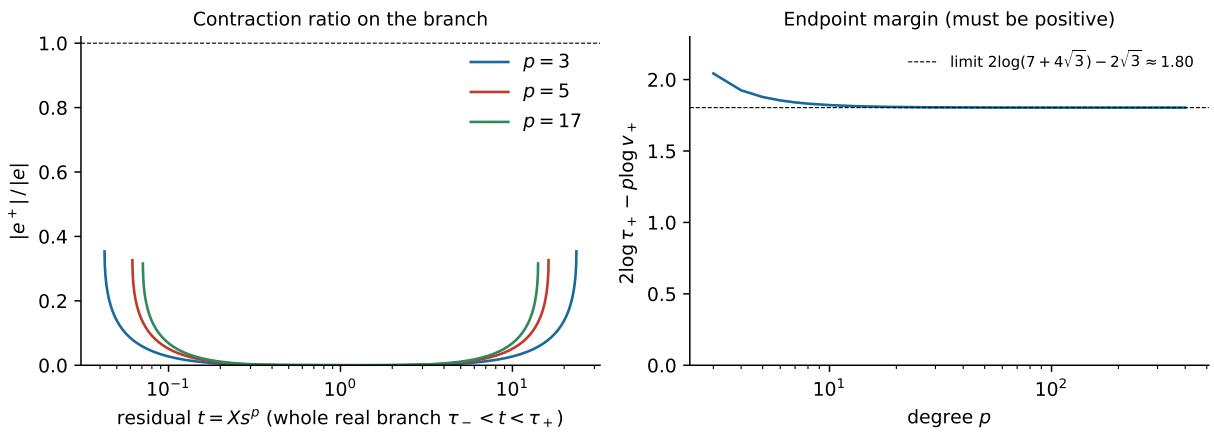


Figure 2: Sampled contraction ratios and endpoint margins illustrate the proved inequalities. The left curves use $p = 3, 5, 17$; the horizontal line is the strict-error threshold one. The right curve samples degrees 3–400. Its limiting margin is $2 \log(7 + 4\sqrt{3}) - 2\sqrt{3} \approx 1.80373$. No uniform numerical bound such as 0.36 is asserted as a theorem.

The whole-band result still requires real branch membership. The next constructions provide globally defined alternatives; Section 6 uses AD to enter the band and to handle excluded finite charts.

4 The five Pandrosion constructions

The fixed circle solves for a correction externally. A second architecture keeps the power endpoint $E = R(s^p)$ and changes the interception support inside the rectangle. Prepare $M = (0, H(1 - X^{-1}))$. The slope of ME is $m = H(1 - t)/(WX)$. A line through A of slope $k_s = H\kappa/(WX)$ meets the parallel to ME through $P = R(s)$ at T_A . The horizontal through T_A gives P^+ , with

$$s^+ = s \frac{k_s}{k_s - m} = s \frac{\kappa}{\kappa - 1 + t}. \quad (6)$$

This one identity explains the four accelerated supports; each is a different device for supplying κ . Here we deliberately use the triangle protocol from Section 2, rather than the rail telescope: horizontals meet the diagonal OB at $B_j = (Ws^j, H(1 - s^j))$ and the left rail at $L_j = L(s^j)$. Parallels to L_1B carry B_j to L_{j+1} , by similar triangles. The p th horizontal gives $E = R(s^p)$.

Reference construction: linear MA. MA provides a baseline, not an additional accelerated method. Write its lower-right endpoint as $A_{\text{lin}} = B$. The parallel to ME through P meets MB ; a horizontal returns the next state. For $s \neq 1$,

$$\mathcal{T}_{\text{MA}}(s) = 1 - \frac{X - 1}{XS_p(s)}, \quad S_p(s) = \sum_{j=0}^{p-1} s^j.$$

For $X > 1$ this scalar map is increasing and $\mathcal{T}_{\text{MA}}(s) - s = (1 - Xs^p)/(XS_p(s))$. It therefore converges monotonically to a from every positive start, with the removable scalar extension at $s = 1$. Its derivative at a is $1 - pa^{p-1}/S_p(a) \in (0, 1)$: convergence is linear.

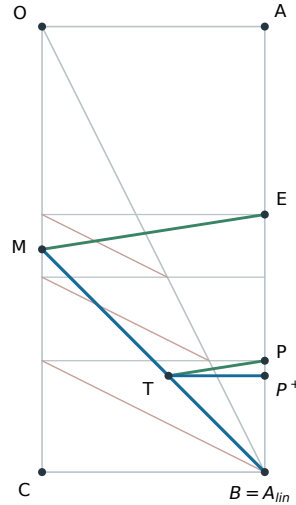


Figure 3: Pandrosion MA, the linear reference construction, at $p = 3, X = 2, s = 3/4$. Thin lines construct the power by similar triangles. The green parallel meets the blue support MA_{lin} ; the horizontal gives P^+ . The historical endpoint A_{lin} is the lower-right vertex B in this paper. At $s = 1$ the geometric intersection is coincident despite the removable scalar extension.

AK: a fixed line. Prepare $K = (W(1 - X/p), 0)$. The support AK has $\kappa = p$, so $\mathcal{T}_{AK}(s) = ps/(p - 1 + t)$, Newton under inversion. It has order two and converges from every positive start.

$$O_{\text{AD}} = \left(W - \frac{2W}{p-1}, H - \frac{H(p+1)}{X(p-1)} \right).$$
$$\mathcal{T}_{\text{AD}}(s) = s \frac{p+1+(p-1)t}{p-1+(p+1)t}. \quad (7)$$

Both drawings preserve the full triangular power stage. The support encodes the correction law: a fixed slope for AK, or a radius-dependent slope for AD. The green transport and horizontal readout are otherwise identical.

4.2 Projective AD: order four

Two joins supply the support. Prepare

$$\ell : y = H \left(1 + \frac{5p-1}{(p+1)X} \right), \quad Z_{\text{proj}} = \left(W \left(1 + \frac{5p-1}{2p(2p-1)} \right), H \left(1 + \frac{p+1}{(2p-1)X} \right) \right).$$

Set $D_{\text{proj}} = Z_{\text{proj}}E \cap \ell$ and join AD_{proj} . Substitution in (6) gives

$$\mathcal{T}_{\text{proj}}(s) = s \frac{2p((2p-1)t + p + 1)}{(p+1)t^2 + 2(2p-1)(p+1)t + (2p-1)(p-1)}. \quad (8)$$

The direct-root correction is Padé [2/1]; the reciprocal-state factor displayed here is [1/2]. Denote its positive denominator by $Q_{\text{proj}}(t)$. Direct differentiation gives

$$\mathcal{T}'_{\text{proj}}(s) = -\frac{2p(p^2-1)(2p-1)(t-1)^3}{Q_{\text{proj}}(t)^2}.$$

The map has its global maximum a at $s = a$. Also $\mathcal{T}_{\text{proj}}(s)/s - 1 = (1-t)((p+1)t + 5p-1)/Q_{\text{proj}}(t)$. Thus a first step lands at or below a , and subsequent steps increase to a . Integrating the derivative at a gives exact order four. The archival companion gives further dynamics and the original triangle-chain realization [11].

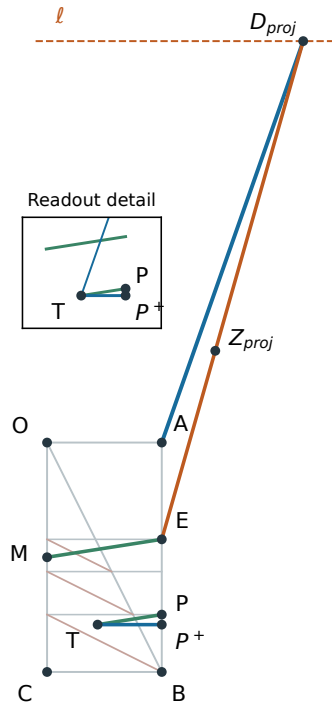


Figure 6: Projective AD [2/1], order four, at the same cubic state. The orange line $Z_{\text{proj}}E$ meets the prepared horizontal ℓ at D_{proj} . The blue support AD_{proj} receives the green correction. The inset separates P , T and P^+ .

The fifth construction. Displacing the arc center from the prepared vertical makes the height a genuine square root. The decentered AD support is developed next, because its global fifth-order behavior changes the choice of device rather than only its coefficients.

4.3 A decentered arc with a global fifth-order map

For $p > 2$, define

$$q(t) = \sqrt{\frac{(p-2)(t^2+1) + (10p+4)t}{12p}}, \quad g(t) = (p-1)q(t), \quad \mathcal{T}_{\text{arc}}(s) = s \frac{1+g(t)}{t+g(t)}. \quad (9)$$

For integer $p \geq 3$, put $\eta = (p-1)\sqrt{(p-2)/(12p)}$, $\zeta = (5p+2)/(p-2)$, $d_0 = W/\eta$ and $\delta = (H/X)\sqrt{\zeta^2-1}$. Prepare

$$G_0 = (W, H + H\zeta/X), \quad O_{\text{arc}} = (W - d_0 + \delta, H - H/(X\eta)), \quad x = W - d_0.$$

Draw the arc centered at O_{arc} with radius $EG_0 = (H/X)(t + \zeta)$ and take the lower intersection D_{arc} with the prepared vertical. Its vertical distance below the center is

$$\sqrt{EG_0^2 - \delta^2} = \frac{H}{X} \sqrt{t^2 + 2\zeta t + 1}.$$

The support AD_{arc} consequently gives $\kappa = 1 + g(t)$ in (6). All preparation parameters use rational operations and square roots of positive quantities. The radicand stays positive for $t > 0$.

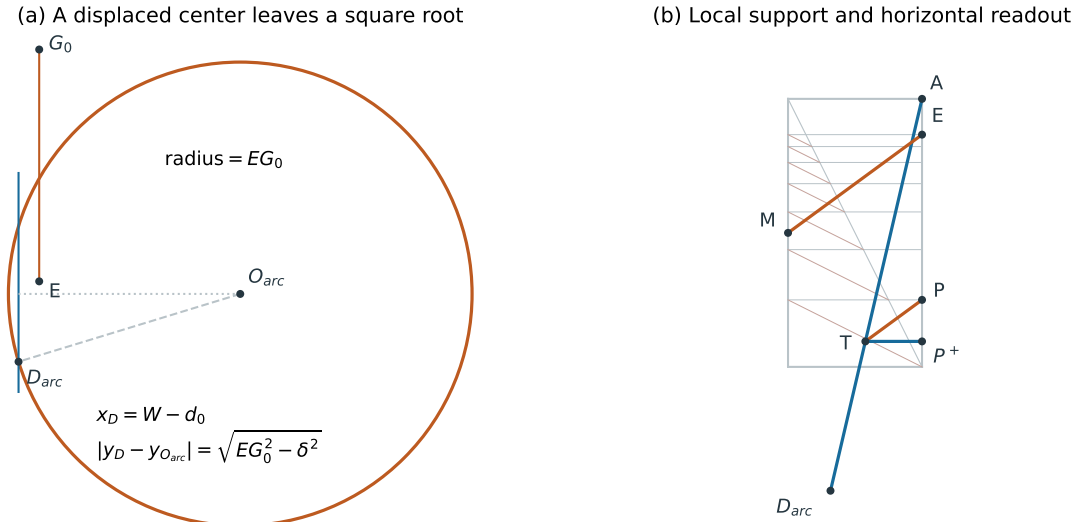


Figure 7: The decentered-arc device for $p = 7, X = W = 2, H = 4, s = 3/4$. The full circle geometry is shown separately from the local report, at independently chosen equal-axis scales. The displacement δ makes the vertical circle intersection perform a square root. The local panel retains the complete support and distinguishes the reported state.

Unlike the inverse-circle correction, this map is defined for every positive residual and requires no choice between two inverse-model branches. It uses a mobile radius at every step. Unlike rational diagonal Padé, it realizes the correction with a single radical rather than two rational factors. These are practical differences in geometric realization, even though all three maps have fifth local order.

Centered AD and decentered AD use the same mobile support operations: one arc and one join. Their fixed preparations and coordinate scales differ. This is the precise sense in which order five is obtained at the mobile support cost of order three.

4.4 Global monotone convergence of the decentered map

Theorem 4.1. *For real $p > 2$, $X > 0$ and every $s_0 > 0$, (9) converges to a monotonically without crossing it. Its logarithmic error satisfies*

$$e^+ = \frac{(p-2)(p^2-1)(3p+1)}{1440}e^5 + O(e^7). \quad (10)$$

Proof. The radicand is positive and $q(1/t) = q(t)/t$, so the correction is reciprocal. Write

$$\mathcal{E}(t) = \frac{\log t}{p} + \log(1 + g(t)) - \log(t + g(t)).$$

Define the auxiliary polynomials

$$\begin{aligned} A_*(t) &= (p-1)(t^2+1) + (10p+2)t, \\ B_*(t) &= (t+1)(6pt - (t-1)^2), \\ K_*(t) &= (p^2-5p-2)(t-1)^2 + 12p(3p+1)t. \end{aligned}$$

Differentiation and factorization give

$$\frac{d\mathcal{E}}{d \log t} = \frac{(p-2)(p-1)^2(qA_* - B_*)}{12p^2g(1+g)(t+g)}, \quad (11)$$

$$q^2A_*^2 - B_*^2 = \frac{p+1}{12p}(t-1)^4K_*. \quad (12)$$

Here $A_* > 0$. If $B_* \leq 0$, then $qA_* - B_* > 0$. Otherwise $(t-1)^2 < 6pt$. When $p^2 - 5p - 2 \geq 0$, $K_* > 0$ directly; otherwise

$$K_* = 6p^2(p+1)t + (p^2-5p-2)((t-1)^2 - 6pt) > 0.$$

Equation (12) now gives $qA_* > B_*$ for $t \neq 1$. Therefore $\mathcal{E}$ is strictly increasing with $\mathcal{E}(1) = 0$: the new logarithmic error has the old sign. Meanwhile $\Phi(t) - 1 = (1-t)/(t+g)$ moves the state toward the root. The orbit is monotone and bounded between s_0 and a . Its positive limit is a fixed point by continuity, and $\Phi(t) = 1$ implies $t = 1$.

Near one, rationalize $qA_* - B_*$ using (12). Since $q = 1$, $g = p-1$, $A_* = B_* = 12p$ and $K_* = 12p(3p+1)$ at $t = 1$, this gives

$$\lim_{t \rightarrow 1} \frac{d\mathcal{E}/d \log t}{(t-1)^4} = \frac{(p-2)(p^2-1)(3p+1)}{288p^5}.$$

Substituting $t = e^{pe}$ and integrating yields (10); reciprocity makes the analytic logarithmic map odd. $\square$

At $p = 3$, the leading coefficient is $+1/18$, whereas the fixed-circle inverse has $-1/18$. The former stays on the starting side; the latter alternates near the root. At $p = 2$, both scalar radical formulas reduce to the exact square-root correction, but the displayed decentered preparation must be replaced by a mean-proportional construction.

5 What the shorter construction saves

A useful comparison holds the state, requested accuracy and counting convention fixed. The rational competitors can be built on the same rail telescope. A prepared projector

$$Z(\mu, \nu) = \left(-\frac{W\mu}{1-\mu}, H\left(1 - \frac{\nu}{1-\mu}\right) \right)$$

sends $Q = R(q_p)$ to $\mathcal{U}_{f(q_p)}$, with $f(q) = \mu + \nu/q$. Two such projections and two rail joins multiply a stored value by f/g : a four-join module.

For Halley choose $f = (p-1)/(4p) + (p+1)/(4pt)$ and $g = (p+1)/(4p) + (p-1)/(4pt)$. Since $t = X2^{p-1}q_p$, each is of the required projector form. The cost is $2p-1+4 = 2p+3$ joins.

For rational diagonal Padé set $a_5 = (p-1)(2p-1)$, $b_5 = 8p^2-2$, $c_5 = (p+1)(2p+1)$. The correction is

$$\Phi_{\text{rat}}(t) = \frac{a_5 t^2 + b_5 t + c_5}{c_5 t^2 + b_5 t + a_5} = \prod_{r \in \{r_+, r_-\}} \frac{t+r}{rt+1}, \quad r_{\pm} = \frac{b_5 \pm \sqrt{12p^2(4p^2-1)}}{2a_5}.$$

Each factor uses the four-join module with $f_r = [1+r/t]/[2(1+r)]$ and $g_r = [r+1/t]/[2(1+r)]$. Both reuse Q , giving $2p+7$ joins. Its scalar map is globally monotone: its derivative is $a_5 c_5 (t-1)^4 / (c_5 t^2 + b_5 t + a_5)^2$, and numerator minus denominator is $6p(1-t)(1+t)$.

Table 1: Generic mobile protocol costs; fixed preparation and final inversion are excluded. A parallel macro is $2J + 3C_{\text{arc}}$. The scope column describes the exact scalar map.

Construction	Order	J	$P_{\parallel}$	C_{arc}	Scope
MA rectangle	1	2	$2p+1$	0	$X > 1$
AK rectangle	2	2	$2p+1$	0	Global
AD rectangle	3	3	$2p+1$	1	Global
Projective AD	4	4	$2p+1$	0	Global
Decentered AD	5	3	$2p+1$	1	Global
Pencil Halley	3	$2p+3$	0	0	Global
Pencil rational Padé	5	$2p+7$	0	0	Global
Fixed-circle inverse	5	$2p+2$	0	0	Real band
Circle-AD protocols	5	Section 6			Global

The fixed circle saves five joins per regular update relative to this rational fifth-order realization. A reproducible cubic example makes both the gain and the price of control visible.

Table 2: Steps / mobile joins to obtain $|s/a - 1| \leq 10^{-d}$ from $p = 3$, $X = 2$, $s_0 = 3/4$, computed at 250 digits. The last two rows use binary residual caching and add 12 initial joins; every circle trial in this example is regular and accepted.

Protocol	$d = 6$	$d = 12$	$d = 30$	$d = 60$
Pencil Halley	2 / 18	3 / 27	3 / 27	4 / 36
Pencil rational Padé	1 / 13	2 / 26	2 / 26	3 / 39
Circle, linear power	1 / 8	2 / 16	2 / 16	3 / 24
Circle, branch entry	1 / 18	2 / 36	2 / 36	3 / 54
Circle, residual gate	1 / 28	2 / 56	2 / 56	3 / 84

These are geometric work counts under the stated convention, not circuit time or energy measurements. Branch entry and the optional residual gate are developed next.

6 From a convergence band to all positive starts

The whole-branch theorem removes the need to test the improvement of a correctly selected, regular circle step. It does not create real intersections outside the band or remove affine chart singularities. We therefore distinguish two global protocols: a branch-driven one using the new theorem, and the residual-controlled V21 protocol whose formal switching proof is independent of it.

6.1 AD entry and geometric exceptions

Cache $E = R(s^p)$ and $R(t)$, and stop if $t = 1$. Attempt the descending circle readout. Use it whenever the crossing is transverse and its readout finite; otherwise use centered AD. Refresh the power and residual of the output. Thus AD supplies entry to the band and any exceptional readouts, rather than checking every circle step's improvement.

Proposition 6.1 (Global branch-driven protocol). *For every integer $p \geq 3$, $X > 0$ and $s_0 > 0$, this exact geometric protocol converges to $a = X^{-1/p}$. With the preparation specified below, it eventually uses the circle step unless already at the root, and has local order five.*

Proof. Outside the real band all steps are AD. Its global convergence brings the residual into the open band after finitely many steps. Inside, the circle strictly reduces $|\log t|$ by Theorem 3.3; AD also strictly reduces it away from the root. Both preserve the band.

Chart exceptions can switch between these maps. They cannot prevent convergence: on the compact symmetric log-residual interval determined by the first entry, both scalar post-error functions are continuous. If the decreasing errors had limit $\ell > 0$, their maximum would be uniformly smaller than the current error on the compact set $\ell \leq |\log t| \leq |\log t_{\text{entry}}|$. Every late step would decrease the error by a fixed positive amount, a contradiction. Hence the error tends to zero.

The finite chart is regular near the root. For the compact preparation ($p \neq 4$), $\rho = 1/2$ and

$$b(1) = \frac{48p(p+4)(p+1)}{(p-1)\Lambda_p} > 0;$$

for $p = 4$, use the separate chart in Appendix A, where direct substitution also gives $\rho \neq 1$ and $b(1) \neq 0$. Geometrically, $v \neq \rho$ keeps BG from being parallel to the top rail, while $b \neq 0$ keeps G distinct from B . At the root $\Delta_p(1) = 36p^2 > 0$, so continuity preserves all three conditions nearby. The eventual update is therefore the order-five circle map. $\square$

This proposition's switching/compactness assembly is a written proof. Lean checks the strict branch inequality and the chart-at-root certificates separately. The fully formal global controller remains the next, more conservative protocol.

6.2 The residual gate remains an independent option

For a positive circle candidate c , construct $u = Xc^p$ and accept precisely when

$$(t \leq 1 \wedge t \leq u^2 \wedge tu^2 \leq 1) \quad \text{or} \quad (1 \leq t \wedge u^2 \leq t \wedge 1 \leq tu^2). \quad (13)$$

Otherwise use AD. Three rail comparisons implement this condition, equivalent to $|\log u| \leq |\log t|/2$. The errors decrease; if their limit were positive, eventually no halving step would be possible, leaving only AD and forcing the limit to zero. This is the V21 global switching theorem, including the incidence-based gate, formalized in `RectangleCircleSafeguard`. The strict branch theorem alone does not assert this stronger half-error inequality. The gate is retained as an independently justified controller, not as a necessary test inside the proved branch.

6.3 Geometric arithmetic and complete trace counts

The controller uses a finite, five-join product. Given $R(a), R(b)$ with $b > 0$, test $R(b) = B$; if equal, reuse $R(a)$. Otherwise $CR(b)$ gives $V = \mathcal{U}_{1/b}$. Project $R(a)$ through $\mathcal{U}_{1/2}$ to $L(a/2)$; its join to V gives $R(ab/2)$. Project through $\mathcal{U}_{1/2}$ and then $\mathcal{U}_{1/4}$ to obtain $L(ab/4)$ and $R(ab)$. The exact unit shortcut handles the otherwise infinite center.

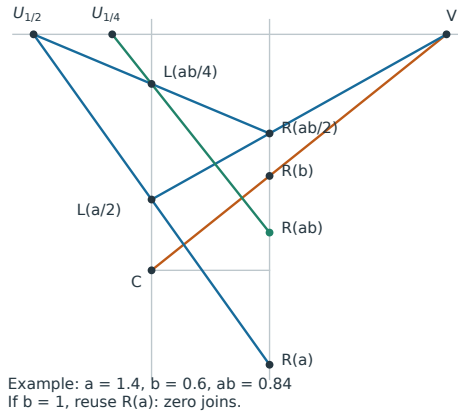
A fixed positive scale d uses two joins, through $\mathcal{U}_{d/[2(1+d)]}$ and $\mathcal{U}_{1/[2(1+d)]}$. Binary powering with $m = \lfloor \log_2 p \rfloor + \text{popcount}(p) - 1$ products therefore constructs $R(s^p)$ and $R(Xs^p)$ in at most $5m+2$ joins, including when $s = 1$. Both residual protocols use this power stage.

Table 3: Upper bounds from a cached current power and residual, including the next cache. Prepared objects, existing intersections and decisions are excluded from trace columns and specified below.

Operation	J	$P_{\parallel}$	C_{arc}
Initial power and residual cache	$5m + 2$	0	0
Branch-driven regular circle step	$5m + 8$	0	0
Branch-driven fallback after a full attempted readout	$5m + 10$	2	1
Residual-gate accepted step	$5m + 18$	0	0
Residual-gate full rejection, AD and refreshed cache	$10m + 22$	2	1

A cached circle trial costs six joins: two for $R(kt)$, one for $L(\rho s)$ and three for ZT , BG and the final readout. AD adds two joins, two parallels and one arc. The optional gate adds two products, at most ten joins, plus three ordered comparisons; a rejected trial also requires a second power/residual evaluation.

Ruler-only product: five joins



Gate on one oriented rail

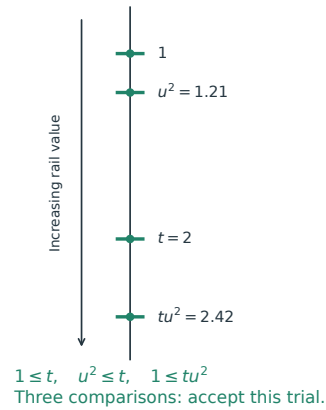


Figure 8: The five-join product and the optional V21 residual gate. The branch-driven V22 protocol reuses the product for powers and scales, but omits the two gate products and its three ordered comparisons. The geometric points and labels are unchanged from the independently checked V21 figure.

For $p = 3$, the cache costs 12 joins; branch-driven regular steps cost at most **18 joins**, and full fallback at most $20J + 2P_{\parallel} + C_{\text{arc}} = 31$ expanded traces. The optional gate costs 28 joins on acceptance and 53 expanded traces after full rejection. At $p = 10^6$, $m = 25$ gives 133 versus 143 joins on acceptance. None of these is a global eight-line bound.

Branch, incidence and root-stop decisions remain explicit. Each power evaluation makes m product identity tests; a completed gate adds two. The trace inventory is checked by instrumented intersections, not a Lean execution-cost semantics. The two 144-trajectory campaigns pass through degree one million. Distant rail points and ill-conditioned near-unit products remain possible: exact global convergence does not imply uniform drawing size or floating-point accuracy.

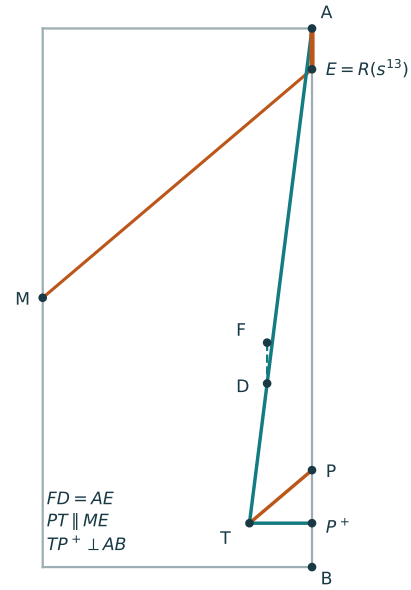
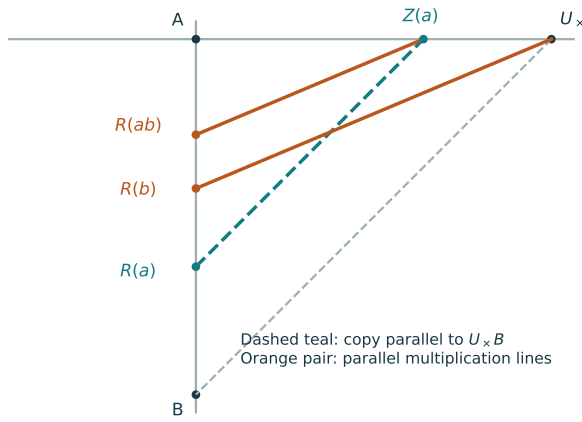
7 Binary geometry and the path to an analog calculator

The linear telescope is attractive for a few degrees. At large p , binary powering gives a smaller arithmetic architecture. Define a top encoding $Z_{\times}(q) = (W + Hq, H)$ and the fixed unit $U_{\times} = (W + H, H)$. A parallel to $U_{\times}B$ copies $R(a)$ to $Z_{\times}(a)$. The parallel through $Z_{\times}(a)$ to $U_{\times}R(b)$ then meets the right rail at $R(ab)$: its slope is b and its horizontal displacement is $-Ha$.

Squaring the accumulator and multiplying by s at each binary one produces $E = R(s^p)$. With $L = \lfloor \log_2 p \rfloor$ and $h = \text{popcount}(p)$, there are $m = L + h - 1$ multiplications. Reusing the top copy of s , the power stage costs $mJ + (m + L)P_{\parallel}$. It feeds any of the four rectangle supports unchanged.

(b) Fast output E feeds the AD support

(a) One geometric multiplication



(c) Binary schedule: $13 = (1101)_2$; five multiplications instead of twelve

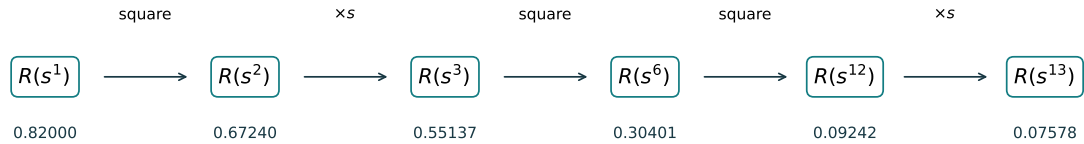


Figure 9: The fast-power construction and its AD readout, adapted from the V30 companion. The case $p = 13 = (1101)_2$ uses five products along exponents 1, 2, 3, 6, 12, 13. A product is an actual pair of line incidences, not a numerically inserted power. The output E enters the same moving support and horizontal readout.

The construction is an operation graph for a circuit: it identifies which signals must exist before a cell can act, and which intermediate values must survive a later correction. The implementation obligations are concrete:

Geometric object	Electrical role	Property to measure
Rail coordinate	Scaled voltage or current	Dynamic range and offset
Product incidence	Multiplier and weighted sums	Gain error and settling
Circle crossing	Quadratic solve plus branch choice	Stability and branch margin
Retained point	Sample-and-hold memory	Leakage and sampling error

The next page separates the existing AD circuit from the quadratic cell suggested by the fixed-circle construction.

7.1 From an incidence equation to an analog cell

What has been implemented. The V30 prototype realizes centered AD, of order three. A digitally prepared scale c makes $Y \approx Xc^p \in [1, 2]$, with $s = cu$. Storing $u \approx 1$ directly would give sensitivity $\partial \log(Yu^p)/\partial u = p/u$. Instead store $q = p(u-1)$: the sensitivity becomes $1/u$. Centered binary products evaluate the residual, followed by

$$q^+ = q + \frac{2(1+q/p)(1-t)}{1+t+(t-1)/p}, \quad \hat{r} = \frac{1}{c(1+q/p)}.$$

The V30 companion gives the exact centered product recurrences, netlists and calibration protocol [12]. There are 25 power products at $p = 10^6$, and at most 37 over the programmed range $3 \leq p \leq 10^6$.

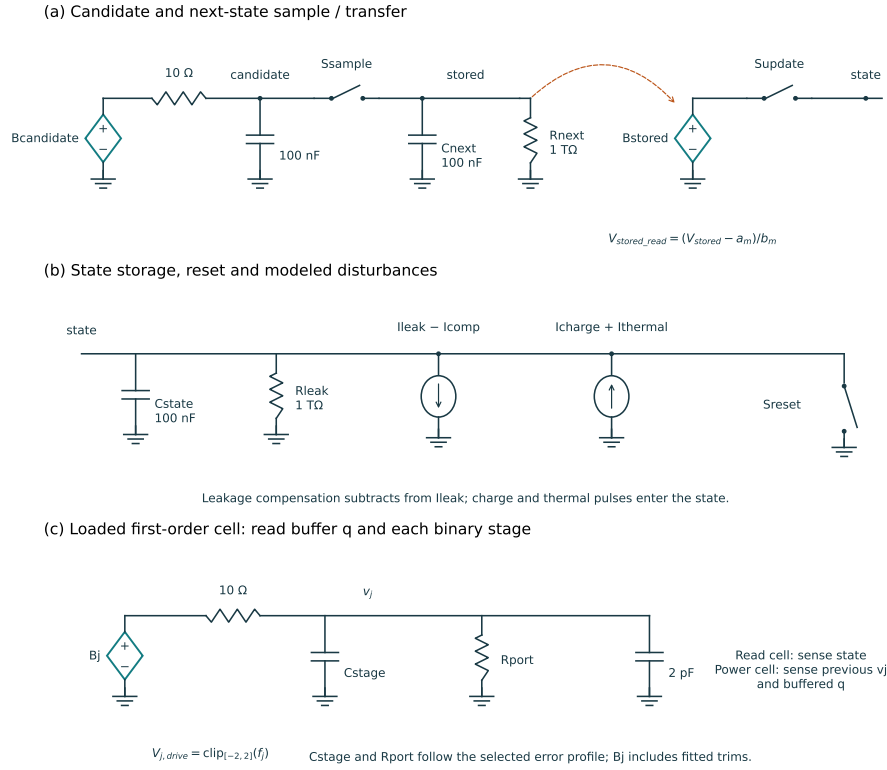


Figure 10: Existing AD behavioral topology from V30. State memory, loading and switches are explicit; diamonds are behavioral arithmetic sources. The drawing identifies where state must settle and be retained. It is the implemented order-three baseline, not an electrical implementation of the circle solver.

What the fixed circle specifies for a new cell. Its geometric incidence is an exact circuit specification: solve $N(v) - tD(v) = 0$ on the branch through $v = 1$, then form $s^+ = sv$. A circle on paper does not by itself supply an electrical component. A candidate realization would use multipliers, weighted sums and feedback to enforce this quadratic equation, with branch selection and AD fallback.

As with AD, the useful electrical variable at large p is centered: put $z = p(v-1)$. Dividing the incidence equation by 12 gives the exact relation

$$0 = (1-t) - \frac{(1-2/p) + (1+2/p)t}{2} z + \frac{(1-3/p+2/p^2) - t(1+3/p+2/p^2)}{12} z^2.$$

At $(t, z) = (1, 0)$ the derivative of the left side with respect to z is -1 , independently of p . This removes the trivial large-degree scaling from the local quadratic cell. At a branch endpoint that derivative vanishes: an electrical design must restrict its operating margin or fall back before the tangency. Algebraic centering alone does not control noise or feedback settling.

7.2 What the prototype teaches about the construction

The existing evidence answers three distinct design questions. The 8,243-pair arithmetic campaign checks that one programmable binary schedule covers many degrees and input scales. The calibrated behavioral SPICE study meets one ppm decoded-root error in 64 cases, with worst error 0.279 ppm at 23.985 ms: memory, calibration and readout determine a practical error floor even when the iteration is exact. A 4.785 ms policy meets one ppm on 18 new cases, demonstrating a precision–time tradeoff within that same behavioral model, not a measured silicon speed.

Preparation can dominate an apparently impressive large-degree example. At $p = 10^6$, $X = 500000$, the digitally prepared $1/c$ is already within about 2.477×10^{-7} relative error of the root. The analog loop refines it. A fair comparison must therefore include preparation, programming and conversion; the recorded circuit readout times exclude those operations.

For the proposed circle cell, the next experiment is consequently specific: retain the same power module and memories, replace the AD correction by the centered quadratic relation, and measure wrong-branch events, settling time and decoded-root error under the same loading and noise assumptions. Fifth-order mathematical convergence can reduce the number of updates; the extra nonlinear solve may cost time and energy per update. The geometry supplies an exact operation graph and testable cell equations. A performance advantage requires that electrical comparison, which has not yet been performed.

8 Evidence, scope and reproducibility

Mathematical scope. The strict-error inequality over the whole inverse branch is formally proved for integer $p \geq 3$. Invariant-band convergence and the branch-driven global extension are written consequences. The residual-gated circle–AD controller has a separate formal global convergence proof. The concrete inverse-circle local-order assembly and the decentered map’s global monotone convergence are written proofs. The linear MA argument follows directly from its increasing scalar map; the finite diagram excludes its coincident state. The rational baselines belong to classical iteration families [13–15]. The explicit arrangements and counted protocols are the objects proposed here; their historical priority and optimality remain open. Iterative approximation of a cube root is distinct from its exact finite straightedge-and-compass construction.

Formal scope. Lean and mathlib [16, 17] serve as algebraic and incidence checks, with additional analytic theorems where the modules actually export them. The relevant modules are:

Module family	What the checked development supplies
RectangleProjective	Projection/readout identities and order-four dynamics.
RectanglePencils*	Rail operations, rational projectors, dynamics and order results.
RectangleFixedCircleBranch	Whole-branch strict error contraction and endpoint bounds (33 audited declarations).
RectangleFixedCircle	Quadratic identities, radical charts, preparation certificates, readout and a parameterized local limit.
RectangleDecenteredArc	Polynomial certificates used in the written radical proof.
RectangleCircleSafeguard	Geometric residual gate, ruler-only products and powers, global hybrid convergence and local transfer.
RectangleFastPower	Product incidence and geometric binary recursion.
AnalogFastAD	Six exact centered-arithmetic identities.

The full radical convergence argument, inverse-branch and band/entry convergence assemblies, and physical trace count are written arguments. The sampled ratios below 0.36 do not provide a formal half-error bound. Neither Lean nor the numerical checks certify the browser renderer, device noise, ADC implementation or fabricated silicon. The named audits restrict transitive axioms to the standard propext, Classical.choice and Quot.sound.

8.1 Reproduction

The [V22 repository guide](#) links sources, proofs and verification commands. The [interactive gallery](#) shows planar constructions and their optional spherical display.

Independent intersections, symbolic identities and scalar comparisons check 162 circle configurations at 250 digits, 108 arc configurations at 160 digits, and 144 residual-gated and 144 branch-driven trajectories at 180 digits. The branch and safeguard modules have separate axiom audits. Frozen V20 and analog V30 packages retain detailed antecedents and circuit evidence.

Citation metadata accompany this version. The paper source, editable figures and datasets are versioned, and the PDF is rebuilt by CI. The repository includes an MIT LICENSE; V21 and its safeguard source are retained as a reproducible archive.

9 Conclusion

A rationally prepared circle realizes a fifth-order correction in $2p + 2$ joins, and its selected inverse branch strictly contracts logarithmic error throughout the real band. AD entry and chart fallback extend the algorithm to arbitrary positive starts, with a separately counted cost. The five Pandrosion supports and rational pencil baselines show how geometry redistributes the arithmetic work.

The same construction language specifies analog cells and their dependencies. Centering the circle's quadratic relation gives a degree-independent derivative at the root; implementing and characterizing that cell remains the next electrical task. The existing centered AD prototype supplies a reproducible baseline. Better-conditioned charts, lower control costs and a measured quadratic-cell error/settling budget are the remaining priorities.

A Rational preparation for every integer degree

This appendix makes the general-circle protocol reproducible without searching through an earlier manuscript. Write $v = p - 3$ and use α_p, γ_p from (3). Define

$$\begin{aligned}\mathcal{D}_p &= 625v^5 + 10525v^4 + 56797v^3 + 88639v^2 + 14758v + 664, \\ \mathcal{K}_p &= 625v^5 + 1975v^4 + 4147v^3 + 2905v^2 + 448v + 340, \\ \chi_p &= 288p(p-2)(25p^2-244)/\mathcal{D}_p, \\ u_p &= 5 - 4\alpha_p/(9\gamma_p), \quad w_p = 8(5-2p)/(9(p-1)).\end{aligned}$$

Take

$$\begin{aligned}\rho &= \frac{1}{3}, & k &= \mathcal{K}_p/\mathcal{D}_p, & j &= -3\chi_p w_p/4, \\ h &= 2 + 2j - \chi_p u_p/2, & z_x &= 2 - \chi_p, & z_y &= \frac{384p(25p^3 - 50p^2 - 415p + 974)}{\mathcal{D}_p}.\end{aligned}$$

Then the incidence certificate in Theorem 3.2 has multiplier $\lambda_p = -64k\chi_p/(9\gamma_p)$. The positive coefficients imply $\mathcal{D}_p, \mathcal{K}_p > 0$ for $v \geq 0$; $25p^2 \neq 244$ for integer $p \geq 3$ makes χ_p, λ_p, j nonzero. All parameters are rational. The circle is drawn with center (h, j) through B . This uniform chart is checked symbolically; the Lean preparation certificates cover the compact chart below and the separate degree-four specialization. At $p = 4$ the formulas give the finite values

$$k = 145/2389, \quad O_\Gamma = (13502/7167, 3328/2389), \quad Z = (-214/2389, 2432/2389).$$

The cubic worked example uses a more compact chart. For $p \neq 4$, put $\Lambda_p = p^3 + 12p^2 + 39p - 26$ and take

$$\begin{aligned} \rho &= \frac{1}{2}, & k &= \frac{(p-2)(p^2-4p+13)}{\Lambda_p}, \\ h &= 2 - \frac{72p(p+4)}{(p-1)\Lambda_p}, & j &= \frac{24p(p-2)(p+4)}{(p-1)\Lambda_p}, \\ z_x &= 2 - \frac{24p(p+4)(p-2)}{(p-4)\Lambda_p}, & z_y &= \frac{24p(p^2+2p-26)}{(p-4)\Lambda_p}. \end{aligned}$$

Here $\lambda_p = -384p(p-2)(p+4)(p^2-4p+13)/[(p-4)(p-1)\Lambda_p^2]$. Substitution checks both charts. For numerical evaluation the branch can be written without the removable cancellation at large t :

$$v(t) = \begin{cases} \frac{\alpha_p - t\gamma_p}{\beta_p(1-t) + \sqrt{\Delta_p(t)}}, & t \leq 1, \\ \frac{\sqrt{\Delta_p(t)} - \beta_p(1-t)}{t\alpha_p - \gamma_p}, & t > 1. \end{cases}$$

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